William H. Weiskopf, IV

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Experience & Education

Ginkgo Bioworks: Biosecurity

Boston, Massachusetts

Staff Software Engineer

March 2021 - November 2025

Tech lead for a HIPAA-compliant pathogen surveillance platform - COVID-19 in thousands of K-12 classrooms across 18 states, then multi-pathogen biosurveillance at 11 international airports (126 origin countries) - that processed 15M+ samples and underpinned $733M in revenue.

- Led a multi-year platform evolution from single-pathogen COVID testing to multipathogen biosurveillance - refactoring live production systems while preserving historical data, existing visualizations, and CDC reporting pipelines.
- Designed the Results & Sample Refactor: decomposed a monolithic data entity into three domain objects (sample, procedure, result) with one-to-many relationships - enabling multi-panel testing without combinatorial data duplication.
- Led a team of ~11 as interim team lead through a period of rapid organizational change - running ceremonies, managing stakeholders, and keeping engineers focused on delivery; transitioned to a dedicated tech lead role once a manager was hired.
- DevOps: rebuilt CI/CD pipelines around Docker with GitHub Actions, codified infrastructure in Terraform (Auth0, AWS, Aptible, supporting services), migrated off legacy vendor systems, and achieved 8x build performance on self-hosted runners.
- Designed PHI removal from the platform as the business shifted from individual diagnostics to aggregate biosurveillance - balancing regulatory compliance, data retention, and system simplification.

Flask · SQLAlchemy · PostgreSQL · Terraform · GitHub Actions · Docker · Nix · AWS EC2/S3/CloudFront · Auth0 · Aptible · Grafana · New Relic · Cursor

Oliver Wyman: Digital

Boston, Massachusetts

Senior Software Engineer

October 2017 - February 2021

- DevOps: wrote idiomatic containers, stood up CI/CD into Kubernetes across Digital's project portfolio.
- Webservices: REST and GraphQL backends for analytics; built a system cataloguing all of Digital's projects and assets for InfoSec auditability.
- Data infrastructure: designed a stack for consulting teams: ETL, workflow scheduling, schema management, and computational scaling. Built out a migration-based schema framework for a project with a major grocery retailer, eliminating database down-time.
- Sustaining: led long-term operations for five projects in the mature phase of their lifecycle.

Python · Kubernetes & Kustomize · Jenkins Pipelines · Prefect · Node.js (Express & Vue) · Alembic · MS SQL Server · PostgreSQL · GraphQL

Lockheed Martin: Space Systems

Littleton, Colorado

Software Engineer - Orion: NASA's next manned spacecraft

September 2014 - September 2017

- Principal architect in moving Orion ground software from pen and paper to full test automation, empowering NASA requirement verification.
- Built a Python/LaTeX-based engine for creating NASA deliverables, cutting months of paperwork for engineers (per deliverable) to a day.

Python packaging · Pytest · LaTeX · Sphinx · Linux · Vim · Perforce · JIRA · Jenkins · Agile · Test-driven development · Elastic stack

Colorado School of Mines B.S. Engineering, Mechanical Emphasis – CS & Economics minors

December 2013

Capstone: NASA Lunabotics Competition: built a regolith mining robot and competed with international teams at the Kennedy Space Center.

Projects

AI-Assisted Engineering

2025 - Present

- Designed a review workflow for agentic development (Pi, Claude Code): a spec per project, a pull request reviewed thread by thread, and offline checks every change must pass. 1,100+ commits across 15 repositories; 4 pull requests merged upstream (including nixpkgs).
- Directed an agent through a phased build of a disc-ripping pipeline: passed a Blu-ray drive through three virtualization layers (Proxmox hypervisor, Talos VM, Kubernetes device plugin & pod), then specified and verified CD, DVD and Blu-ray edge cases across sessions.

Pi · Claude Code · Context engineering · Agent guardrails · Ollama

reelbarrow

2026 - Present

- Building a service that files ripped discs into a video library, using an LLM (Claude) to match Whisper transcripts to episodes. Each automatic filing needs a second, independent signal, because the model can be consistently and confidently wrong.

Python · FastAPI · Claude API · Whisper

Private Infrastructure Cloud

2015 - Present

Architected a GitOps-driven platform from bare metal through immutable Kubernetes, hosting 24+ services, with archive-class file integrity.

OpenTofu/Terraform · Prometheus/Loki/Grafana · Flux CD · Nix · SOPS secrets · ZFS · TLS · Talos Linux · Proxmox · CoreDNS · Tailscale · OpenEBS